HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Semester 1 - 2016/17
Duration: 2 hours

Question	Marks
1	17
2	8
3	13
4	10
5	14
6	9
7	14
Total Marks	85

Candidates must attempt ALL SEVEN questions.
To receive full credit you must show your work and explain your answers.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

1. Consider the function

$$f_X(x) = \begin{cases} x, & \text{if } 0 \leq x \leq 1, \\ x-1, & \text{if } 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of the function f_X and explain why it is a density function for a random variable X. Give reasons for your answers. [3 marks]

(b) Determine ImX, the image of the random variable X and find $E(X)$, the expected value of X. [3 marks]

(c) Find F_X, the cumulative distribution function for X. [4 marks]

(d) Suppose that $Y = X^2$.

i) Explain why $ImY = [0, 4]$ and show that for $0 \leq y \leq 4$

$$F_Y(y) = F_X(\sqrt{y})$$

where F_Y is the cumulative distribution function of Y. [3 marks]

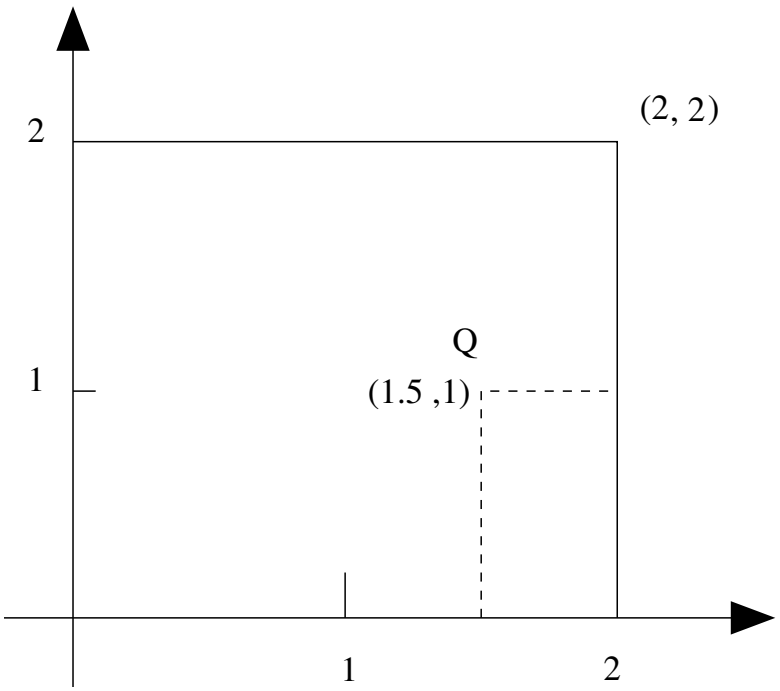
ii) Use part (d)(i) above to show that the density for Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2}, & \text{if } 0 < y \leq 1, \\ \frac{1}{2} - \frac{1}{2\sqrt{y}}, & \text{if } 1 < y \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

[4 marks]

[Total 17 marks]

2. A point Q is chosen uniformly and at random in the square with sides of length 2 and vertices (0,0), (0,2), (2,2), and (2,0). Let X be the distance between the point Q and the nearest point on the boundary of the square. So, for example, if $Q = (1.5, 1)$ then $X = 0.5$.



(a) Using a geometric argument, or otherwise, determine $\mathbf{P}(X > 1/2)$. [2 marks]

(b) Show that the distribution function for X is given by

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0, \\ 1 - (1-x)^2 & \text{if } 0 \leq x < 1, \\ 1, & \text{if } x \geq 1. \end{cases}$$

Hint: For $0 \leq x < 1$, consider $\mathbf{P}(X > x)$. [4 marks]

(c) Determine $E(X)$, the expected value of X. [2 marks]

[Total 8 marks]

[PLEASE TURN OVER]

3. Suppose that X and Y are non-negative random variables with joint density

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{5}(2x+y), & 0 \le x \le 2,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the region where $f_{X,Y}(x,y) > 0$. [1 mark]

(b) Determine $\mathbf{P}(X < Y)$. [3 marks]

(c) Show that the marginal density $f_X(x)$ for X is

$$f_X(x) = \begin{cases} \frac{2x}{5} + \frac{1}{10}, & \text{for } 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

[3 marks]

(d) **i)** Determine the conditional density function $f_{Y|X}(y|x)$ of Y given $X = x$. [2 marks]

ii) Calculate $\mathbf{P}\left(Y < \frac{1}{2} \mid X = 1\right)$. [2 marks]

iii) Determine $\mathbf{E}\left(Y|X=1\right)$. [2 marks]

[Total 13 marks]

4. Suppose that X and Y are *independent* random variables, each with an $Exponential(1)$ distribution.

(a) Let $Z = 3X + Y$. Use the properties of expectation and variance to compute $\mathbf{E}(Z)$ and $Var(Z)$. [4 marks]

(b) Explain why the joint density for the variables X and Y is

$$f_{X,Y}(x,y) = \begin{cases} e^{-x-y}, & \text{for } x > 0, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

[1 mark]

(c) Show that the distribution function $F_Z(z)$ of $Z = 3X + Y$ is given by

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ 1 + \frac{e^{-z}}{2} - \frac{3e^{-z/3}}{2}, & z \ge 0. \end{cases}$$

[5 marks]

[Total 10 marks]

[PLEASE TURN OVER]

5. Suppose that $X_1, X_2, \ ... \ , X_n$ are *independent* random variables with distribution functions $F_1(x), F_2(x), ..., F_n(x)$ respectively. Let $M = \min\{X_1, X_2, \ .... \ , X_n\}$.

(a) Let $F_M(x)$ denote the distribution function for M. Show that for any $x \in \mathbb{R}$,

$$F_M(x) = 1 - (1 - F_1(x))(1 - F_2(x)) \cdots (1 - F_n(x)).$$

[4 marks]

(b) Anne has two alarm clocks which go off each morning, *independently of each other*, at random times which are uniformly distributed between 7 and 8 am. Let X_i for $i = 1, 2$ be the time, measured as minutes past 7 am, at which the ith of the clocks rings (so $X_i \sim U[0, 60]$).

Suppose that Anne gets up as soon as one of the clocks rings and let $T = \min\{X_1, X_2\}$ denote the time, in minutes past 7am, when Anne gets up.

i) Using part (a) above, show that the density $f_T(t)$ for T is given by

$$f_T(t) = \begin{cases} \frac{60-t}{1800}, & \text{for } 0 < t < 60, \\ 0, & \text{otherwise.} \end{cases}$$

[4 marks]

ii) *Given* that Anne gets up at $T = t$ minutes past 7am, it takes her Y minutes to get ready to leave her flat to go university, where Y is uniformly distributed between 0 and 60-t minutes (i.e. *given* that $T = t$, the conditional distribution of Y is $U(0, 60 - t)$). Explain why

$$E(Y \mid T = t) = \frac{60 - t}{2} \qquad \text{for } 0 < t < 60.$$

[1 mark]

iii) Let $W = T + Y$ be the time, in minutes past 7am, that Anne leaves the flat, and use the Partition Theorem, to compute $E(W)$. [5 marks]

[Total 14 marks]

6. **(a)** Suppose that X is a random variable with moment generating function $M_X(\theta)$ and let $Y = a + bX$ for some $a, b \in \mathbb{R}$. Show that $M_Y(\theta) = e^{a\theta} M_X(b\theta)$. [3 marks]

(b) Suppose that $W \sim N(3, 2)$ is a normal random variable.

i) Use $M_W(\theta)$, the moment generating function for W, to compute $E(W^3)$. (Recall that $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$.) [4 marks]

ii) Suppose that $T = e^W$. Use $M_W(\theta)$ to determine $E(T^2)$. [2 marks]

[$Total$ 9 marks]

7. A speed camera is installed on a busy road and a record is kept of each time at which the camera takes a photograph of a speeding car. The times *between* photographs are independent and exponentially distributed with *mean* equal to $\frac{1}{4}$ week.

Starting at $t = 0$, let X_1 denote the time until the first photograph is taken, let X_2 denote the time *between* the first and second photograph, etc. and, in general, let X_n denote the time between the $(n-1)^{th}$ and n^{th} photograph. (So $X_1, X_2, ...$ are i.i.d. exponential random variables with mean $\frac{1}{4}$.)

(a) Let $T_{100} = X_1 + X_2 + ... + X_{100}$ denote the time at which the 100th photograph is taken.

i) Use a moment generating argument to determine the *exact* distribution of T_{100}. [4 marks]

ii) State carefully the Central Limit Theorem. [2 marks]

iii) Use the Central Limit Theorem to approximate the probability that fewer than 100 photographs are taken in the first 28 weeks ,
i.e. approximate $\Pr(T_{100} > 28$ weeks$)$. [3 marks]

(b) It has been estimated that, *given* that a car has been photographed speeding, the probability that it was going more than 15 miles per hour above the speed limit
is $p = 0.4$, independently of the speeds of all other cares.

Suppose that 200 cars have been caught speeding during one year.
Let Y denote the number of these cars that were going more than 15 miles per hour above the speed limit.

i) State the exact distribution of Y. [1 mark]

ii) Use the Central Limit Theorem to determine the (approximate) probability that at most 90 of the 200 cars caught speeding were travelling at more than 15 miles per hour above the speed limit. [4 marks]

[$Total$ 14 marks]

[END OF PAPER]